

ORIGINAL ARTICLE

# Multi-drug resistant, biofilm-producing high-risk clonal lineage of *Klebsiella* in companion and household animals

A. Banerjee<sup>1</sup>, K. Batabyal<sup>1</sup>, A.D. Singh<sup>2</sup>, S.N. Joardar<sup>1</sup>, S. Dey<sup>1</sup>, D.P. Isore<sup>1</sup>, T.K. Sar<sup>3</sup>, T.K. Dutta<sup>4</sup> , S. Bandyopadhyay<sup>5</sup>  and I. Samanta<sup>1</sup> 

<sup>1</sup> Department of Veterinary Microbiology, West Bengal University of Animal and Fishery Sciences, Kolkata, India

<sup>2</sup> Department of Veterinary Public Health, West Bengal University of Animal and Fishery Sciences, Kolkata, West Bengal, India

<sup>3</sup> Department of Veterinary Pharmacology and Toxicology, West Bengal University of Animal and Fishery Sciences, Kolkata, India

<sup>4</sup> Department of Veterinary Microbiology, Central Agricultural University, Aizawl, India

<sup>5</sup> ICAR-Indian Veterinary Research Institute-Eastern Regional Station, Kolkata, India

**Significance and Impact of the Study:** Characterization of antimicrobial resistance is an utmost necessity in animals having direct contact with human as a part of the one health approach. This is the first large-scale monitoring of ESBL/biofilm-producing *Klebsiella* in companion/household animals in India sharing human habitat. Majority of the *Klebsiella* possessed *bla*<sub>CTX-M-15</sub> which is associated with infection in both humans and animals throughout the world. Few nucleotide sequences of *bla*<sub>CTX-M</sub> in the present study are reported for the first time from the companion animals. The emergence of resistance determinants in *Klebsiella* isolated from companion and household animals was correlated with therapeutic antibiotic exposure and contaminated environment, respectively.

## Keywords

companion animals, ESBL, goats, *Klebsiella*, sheep.

## Correspondence

Indranil Samanta, Department of Veterinary Microbiology, West Bengal University of Animal and Fishery Sciences, 37, K.B. Sarani, Belgachia, Kolkata 700037, West Bengal, India.  
E-mail: isamanta76@gmail.com;  
drisamanta@wbuafl.ac.in

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## Abstract

Antimicrobial resistance is a global emergency which needs one health approach to address. The present study was conducted to detect the prevalence of beta-lactamase and biofilm-producing *Klebsiella* strains in rectal swabs ( $n = 624$ ) collected from healthy dogs, cats, sheep and goats reared as companion or household animals in India. The dogs and cats were frequently exposed to third- or fourth-generation cephalosporins for therapy. The sheep and goats were occasionally exposed to antibiotics and had environmental exposure. Phenotypical ESBL ( $n = 93$ ) and ACBL ( $n = 88$ )-producing *Klebsiella* were isolated significantly more ( $P < 0.05$ ) from companion animals than household animals. Majority of the *Klebsiella* possessed *bla*<sub>CTX-M-15</sub>. The sequences *bla*<sub>CTX-M-15.2</sub>, *bla*<sub>CTX-M-197</sub> and *bla*<sub>CTX-M-225</sub> are reported first time from the companion animals. All ACBL-producing isolates possessed *bla*<sub>AmpC</sub>. The present study detected 65.8% of *Klebsiella* strains as biofilm producers possessing the studied biofilm associated genes. The isolates showed phenotypical resistance against chloramphenicol, tetracycline, doxycycline, cotrimoxazole, ampicillin, cefotaxime/clavulanic acid. The present study showed that companion and household animals (dogs, cats, sheep, goats) may act as a carrier of ESBL/biofilm-producing, multi-drug resistant, high-risk clonal lineage of *Klebsiella*.

## Introduction

*Klebsiella* causes opportunistic nosocomial infections in human, and mastitis, pyometra, cystitis, upper respiratory tract infections and enteritis in cattle, goats and

companion animals (Roberts *et al.* 2000; Abdalhamed *et al.* 2018). The bacterial strains can colonize the environmental fresh water including surface water, sewage, soil and mucosal surfaces of gastro-intestinal or respiratory tract of healthy animals. The colonizers of healthy

companion and household animals can act as a source of antimicrobial resistance determinants for the commensals present in the animal owners sharing the common habitats (Guardabassi *et al.* 2004).

The dogs and cats are kept as companion animals in Indian urban and peri-urban localities. The rural communities prefer goats and sheep as household animals as a source of additional income, woman empowerment and animal protein sources like other low- and middle-income group of countries. The companion animals are kept mostly indoor with close contact with the animal owners. The household animals are reared by semi-intensive system which allows roaming around the house during daytime. The household animals are exposed to the outdoor environment with uncontrolled access to human cooking rooms, utensils, drinking water and food prepared for consumption (Day 2019).

Extended spectrum beta-lactamases (ESBLs) are resistance determinants transmitted through horizontal gene transfer in Enterobacteriaceae group of commensal or pathogens frequently in *Klebsiella pneumoniae*, *Escherichia coli* and *Salmonella* (Faith *et al.* 2003). The production of the enzyme is associated with resistance to a variety of  $\beta$ -lactam antibiotics, including penicillins, second-, third- and fourth-generation cephalosporins and monobactams (e.g. aztreonam), but usually not the carbapenems or the cephamycins (e.g. ceftiofur). Moreover, AmpC  $\beta$ -lactamase-producing organisms (ACBL) can produce resistance against cephalosporins, penicillins, cephamycins, monobactams in addition to  $\beta$ -lactamase inhibitors such as clavulanic acid. There are three classical ESBLs, that is, TEM (except TEM-1), SHV (except SHV-1 and 2) and CTX-M (EFSA Panel on Biological Hazards 2011). The CTX-M is observed as the most prevalent type in clinically infected humans throughout the world (Carattoli 2013). ESBL/AmpC-producing Enterobacteriaceae are less reported from companion/household animals than the food animals and are mostly reported in clinical samples (Dierikx *et al.* 2012; Dahmen *et al.* 2013) than the healthy animals (Hordijk *et al.* 2013). Moreover, characterization of ESBL/AmpC-producing *Klebsiella* from companion animals is addressed scarcely (Poirel *et al.* 2013; Donati *et al.* 2014).

Biofilms are microbial communities adhered on biotic or abiotic surfaces and act as a major site for horizontal gene transfer (Hall and Mah 2017). The phenotypic transformation from planktonic cells to immobile stage is a coordinated process regulated by environment and putative factors such as *rpoS*, *sdiA*, *rcsA* and others (Ranjit and Young 2013). *Klebsiella pneumoniae* isolated from clinical samples, wound, catheter, endotracheal tubes of human patients were detected as biofilm producers (Yinon *et al.* 1996; Yang and Zhang 2008; Singhai *et al.*

2012). Biofilm forming *K. pneumoniae* strains isolated from wound swabs were detected to cause prolonged healing of the experimental wound in a rabbit model which revealed the clinical relevance of the biofilm formation (Seth *et al.* 2012). The biofilm formation capacity of *Klebsiella* strains isolated from healthy companion/household animals is not explored.

The present study was conducted to detect the prevalence of beta-lactamase and biofilm-producing *Klebsiella* strains in healthy dogs, cats, sheep and goats reared as companion or household animals in Kolkata metropolitan and different districts of West Bengal (India). The metropolitan city has substantial numbers of companion animals with an increasing trend and the state has highest goat meat production in the country (GOI 2014; The Telegraph 2015/2015). This is the first large-scale monitoring of ESBL/biofilm-producing *Klebsiella* in healthy companion/household animals in India.

## Results and discussion

In the present study, 117 (117/624, 18.75%) isolates were morphologically and biochemically identified as *Klebsiella* and were confirmed by PCR. The isolation rate was significantly more ( $P < 0.05$ ) in companion animals (dogs, cats) than the household animals (sheep, goats; Table 1). Differences in the sample size between the companion and household animals are the probable reason of the differences in isolation rate. The isolation rate varied between different breeds of dogs and lower isolation of *Klebsiella* was detected in studied Indian breeds (Mongrel, Mudhol Hound, Rajapalayam) than the foreign breeds which revealed more suitability of Indian dog breeds as companion animals (Table 2). Among the *Klebsiella* isolates, 98 strains (98/117, 83.7%) were confirmed as *K. pneumoniae* in PCR (Table 1). The prevalence of *Klebsiella* varied widely (3.5–26%) in healthy dogs throughout the world depending on the diet, household environment and breed (Donati *et al.* 2014; Sharif *et al.* 2017). Lower prevalence (5.6%) of *Klebsiella* was detected in healthy goats in India (Aher *et al.* 2012). Use of selective medium and more than one colony in the current study was the probable reason for higher isolation rate of *Klebsiella*.

In total, 93 *Klebsiella* sp. were identified as phenotypically ESBL producers in double disc synergy test isolated significantly more ( $P < 0.05$ ) from companion animals (19.8%) than household animals (8.01%, Table 1). Higher prevalence of ESBL-producing *Klebsiella* in companion animals was associated with therapeutic exposure to the third-generation cephalosporins which the owners could afford (Table S1). Prevalence of ESBL-producing *Klebsiella* in infected dogs and cats varied between the study locations and antibiotic exposure such as

**Table 1** Beta-lactamase and biofilm-producing *Klebsiella* spp. isolated from companion and household animals in West Bengal, India

| Agro-climatic (districts)          | zones | Animal species/numbers of samples | <i>Klebsiella</i> spp.             | <i>Klebsiella pneumoniae</i>                   | Beta-lactamase-producing <i>Klebsiella</i> spp. | Biofilm-producing <i>Klebsiella</i> spp.     |
|------------------------------------|-------|-----------------------------------|------------------------------------|------------------------------------------------|-------------------------------------------------|----------------------------------------------|
| New Alluvial (Kolkata)             |       | 147 (dogs)                        | 31                                 | 29                                             | 29                                              | 28                                           |
|                                    |       | 61 (cats)                         | 11                                 | 9                                              | 9                                               | 8                                            |
| New Alluvial (Howrah)              |       | 39 (dogs)                         | 10                                 | 8                                              | 9                                               | 7                                            |
|                                    |       | 23 (cats)                         | 5                                  | 4                                              | 4                                               | 4                                            |
| Old Alluvial (Nadia)               |       | 41 (dogs)                         | 12                                 | 11                                             | 11                                              | 9                                            |
|                                    |       | 11 (cats)                         | 4                                  | 3                                              | 2                                               | 3                                            |
| Old Alluvial (Malda)               |       | 27 (dogs)                         | 8                                  | 6                                              | 7                                               | 5                                            |
|                                    |       | 13 (cats)                         | 2                                  | 1                                              | 1                                               | 1                                            |
| Total (companion animals)          |       | 362 (dogs: 254, cats: 108)        | 83 (22.9%)* (dogs: 61, cats: 22)   | 71 (19.6%) <sup>a</sup> (dogs: 54, cats: 17)   | 72 (19.8%) <sup>b</sup> (dogs: 56, cats: 16)    | 65 (17.9%) <sup>c</sup> (dogs: 49, cats: 16) |
| Coastal saline (South 24 Parganas) |       | 43 (sheep)                        | 7                                  | 6                                              | 5                                               | 2                                            |
|                                    |       | 33 (goats)                        | 4                                  | 4                                              | 3                                               | 2                                            |
| Red Laterite (Jhargram)            |       | 12 (sheep)                        | 3                                  | 2                                              | 2                                               | 1                                            |
|                                    |       | 18 (goats)                        | 2                                  | 2                                              | 1                                               | 1                                            |
| New Alluvial (Kolkata)             |       | 27 (goats)                        | 4                                  | 2                                              | 2                                               | 1                                            |
| Old Alluvial (Nadia)               |       | 49 (sheep)                        | 4                                  | 4                                              | 3                                               | 2                                            |
|                                    |       | 34 (goats)                        | 3                                  | 3                                              | 2                                               | 1                                            |
| Old Alluvial (Malda)               |       | 15 (sheep)                        | 5                                  | 3                                              | 2                                               | 1                                            |
|                                    |       | 31 (goats)                        | 2                                  | 1                                              | 1                                               | 1                                            |
| Total (household animals)          |       | 262 (sheep: 119, goats: 143)      | 34 (12.9%)* (sheep: 19, goats: 15) | 27 (10.3%) <sup>a</sup> (sheep: 15, goats: 12) | 21 (8.01%) <sup>b</sup> (sheep: 12, goats: 9)   | 12 (4.58%) <sup>c</sup> (sheep: 6, goats: 6) |
| Grand total                        |       | 624                               | 117 (117/624, 18.75%)              | 98 (98/117, 83.7%)                             | 93 (93/117, 79.4%)                              | 77 (77/117, 65.8%)                           |

\*Differs significantly ( $P < 0.05$ ); <sup>a</sup>Differs significantly ( $P < 0.05$ ); <sup>b</sup>Differs significantly ( $P < 0.05$ ); <sup>c</sup>Differs significantly ( $P < 0.05$ ).

Switzerland (3.1%, Zogg *et al.* 2018), Germany (5.06%, Ewers *et al.* 2014), Italy (21.4%, Donati *et al.* 2014), Taiwan (22.7%, Kuan *et al.* 2016) and Japan (39–42%, Harada *et al.* 2016; Maeyama *et al.* 2018). The finding in the present study in companion animals is even higher than the prevalence of ESBL-producing *Klebsiella* in healthy dogs in France and Nigeria (14%, Poirelet *et al.* 2013; Okpara *et al.* 2018).

Lower prevalence of ESBL-producing *Klebsiella* was detected in household animals (sheep: 10.0%, goats: 6.2%) in the present study although third-generation cephalosporins were not used for therapy or with feed, as the drugs or commercial feed were not affordable by the farmers. Similarly, lower prevalence of ESBL-producing *Klebsiella*/*E. coli* was detected in household sheep and goats (8.7–17.7%) in Nigeria which was not correlated with antimicrobial usage (Okpara *et al.* 2018). Occurrence of ESBL-producing *Klebsiella* in household animals in absence of the direct exposure to the third-generation cephalosporins can be correlated with the exposure to the environmental resistance gene pool. Earlier studies in the same geographical location revealed that the water and soil may act as source of antimicrobial resistant bacteria

to the backyard pigs and ducks (Samanta *et al.* 2015; Banerjee *et al.* 2019). The more detailed genetic analysis exhibited that *bla*<sub>CTX-M</sub> gene of *E. coli* isolated from dogs and cats possessed a unique flanking region originated from environmental gene pool due to exposure to the contaminated environment (Maeyama *et al.* 2018).

Majority of the *Klebsiella* isolates possessed *bla*<sub>CTX-M</sub> (83/117, 70.94%) followed by *bla*<sub>TEM</sub> (79/117, 67.52%) and *bla*<sub>SHV</sub> (11/117, 9.4%). Moreover, the isolates from companion animals (*bla*<sub>CTX-M</sub>: 65, *bla*<sub>TEM</sub>: 63, *bla*<sub>SHV</sub>: 9) possessed comparatively more ESBL genes (statistically non-significant) than household animals (*bla*<sub>CTX-M</sub>: 18, *bla*<sub>TEM</sub>: 16, *bla*<sub>SHV</sub>: 2) associated with therapeutic exposure to the third generation cephalosporins. The selected sequences of the PCR products were compared and found cognate with *bla*<sub>CTX-M-15</sub> (95–97% cognate), *bla*<sub>CTX-M-15.2</sub> (93% cognate), *bla*<sub>CTX-M-197</sub> (99% cognate), *bla*<sub>CTX-M-225</sub> (98% cognate), *bla*<sub>SHV-12</sub> (99% cognate) and *bla*<sub>TEM-1</sub> (99% cognate) in BLAST search. The sequences were published by DDBJ with accession numbers LC485946 (*bla*<sub>CTX-M-197</sub> in dog), LC484201 (*bla*<sub>CTX-M-15</sub> in cat), LC485947 (*bla*<sub>CTX-M-15</sub> in dog), LC486087 (*bla*<sub>CTX-M-15.2</sub> in cat), LC487855 (*bla*<sub>CTX-M-15</sub> in goat), LC487856

**Table 2** Isolation of *Klebsiella* spp. from different breeds of dogs in West Bengal, India

| Breeds                              | Numbers of <i>Klebsiella</i> isolates |
|-------------------------------------|---------------------------------------|
| Labrador Retriever ( <i>n</i> = 33) | 11 (11/33, 33.3%)                     |
| Golden Retriever ( <i>n</i> = 27)   | 4 (14.8%)                             |
| German Shepherd ( <i>n</i> = 17)    | 3 (3/17, 17.6%)                       |
| Beagle ( <i>n</i> = 19)             | 6 (6/19, 31.5%)                       |
| Pug ( <i>n</i> = 31)                | 9 (29.0%)                             |
| Boxer ( <i>n</i> = 5)               | 0                                     |
| Saint Bernard ( <i>n</i> = 7)       | 1 (14.2%)                             |
| Spitz ( <i>n</i> = 29)              | 16 (55.1%)                            |
| Dachshund ( <i>n</i> = 11)          | 2 (18.1%)                             |
| Mongrel ( <i>n</i> = 70)            | 9 (12.8%)                             |
| Mudhol Hound ( <i>n</i> = 3)        | 0                                     |
| Rajapalayam ( <i>n</i> = 2)         | 0                                     |

(*bla*<sub>CTX-M-15</sub> in goat), LC488709 (*bla*<sub>CTX-M-15</sub> in sheep), LC488710 (*bla*<sub>CTX-M-15</sub> in sheep) and LC488711 (*bla*<sub>CTX-M-15</sub> in sheep). Higher frequency of *bla*<sub>CTX-M</sub> than *bla*<sub>SHV</sub>/*bla*<sub>TEM</sub> was also observed earlier in *Klebsiella* strains isolated from healthy companion animals (Poirel *et al.* 2013; Harada *et al.* 2016). Majority of the strains isolated from both companion/household animals possessed *bla*<sub>CTX-M-15</sub> in the present study which is supported with the earlier studies both in healthy or in infected companion/household animals (Poirel *et al.* 2013; Ewers *et al.* 2014; Harada *et al.* 2016; Zogg *et al.* 2018; Okpara *et al.* 2018). The CTX-M-15-producing *K. pneumoniae* (international clone ST15) are associated with nosocomial infection in both human and animal population throughout the world (Haenni *et al.* 2011). The putative factor (CTX-M-15) is considered as major contributor in extended spectrum cephalosporin resistance in human and animals (Pitout and Laupland 2008; Harada *et al.* 2016). *Klebsiella* strains possessing *bla*<sub>CTX-M-15.2</sub>, *bla*<sub>CTX-M-197</sub>, *bla*<sub>CTX-M-225</sub> isolated from dogs and cats are not reported earlier. The nucleotide sequence of *bla*<sub>CTX-M-225</sub> is available in Genbank (MK628510.1) detected in *E. coli* strains isolated from human urinary tract infection in India. Possession of *bla*<sub>TEM-1</sub> and *bla*<sub>SHV-12</sub> along with *bla*<sub>CTX-M-15</sub> indicating the linkage of the ESBL genes in *Klebsiella* strains is supported with earlier findings in healthy/infected companion animals (Haenni *et al.* 2011; Poirel *et al.* 2013; Kuan *et al.* 2016; Zogg *et al.* 2018).

*Klebsiella* (88/117, 75.2%) were detected as ACBL producers with cefoxitin–cloxacillin double disc synergy test and all the isolates possessed *bla*<sub>AmpC</sub> in PCR. The sequences of the PCR products were compared and found 99% cognate with *bla*<sub>AmpC</sub> in BLAST search. The isolates from companion animals (63/83, 75.9%) possessed comparatively more *bla*<sub>AmpC</sub> (statistically non-significant) than household animals (25/34, 73.5%) due to therapeutic

exposure of the companion animals to cefotaxime and ceftazidime (Table S1). In India, earlier studies revealed occurrence of AmpC in *Klebsiella/E. coli* strains isolated from poultry, cattle with mastitis, pig and farm environment, and ducks (Samanta *et al.* 2018; Banerjee *et al.* 2019).

The study detected 77 (77/117, 65.8%) *Klebsiella* strains as biofilm producers (moderate or strong) in phenotypic assay. All the isolates possessed the biofilm-associated genes *csgA*, *rcsA*, *rpoS* and *sdiA* in PCR. Biofilm-producing *Klebsiella* were isolated significantly more ( $P < 0.05$ ) from companion animals (17.9%), especially from the foreign breed dogs, than household animals (4.5%; Table 1). Biofilm formation in bacterial strains is associated with response to external stresses including antibiotics for better survival (Kaplan 2011).

ESBL-producing *Klebsiella* (*n* = 93) isolated from companion/household animals showed phenotypical resistance against chloramphenicol (100%), tetracycline (100%), doxycycline (91.3%), co-trimoxazole (91.3%), ampicillin (88.1%), cefotaxime/clavulanic acid (83.8%), norfloxacin (82.7%) and ampicillin/cloxacillin (81.7%; Table 3). The isolates were found sensitive to imipenem-EDTA (100.0%) and gentamicin (43.0%; Table 3). Similarly, phenotypical resistance to tetracycline, quinolones and chloramphenicol was observed in *Klebsiella* strains isolated from companion and household animals (Haenni *et al.* 2011; Poirel *et al.* 2013; Harada *et al.* 2016; Okpara *et al.* 2018). Resistance to tetracycline, doxycycline and quinolone is associated with direct exposure to the antibiotics as the drugs are commonly used in dogs, cats and goats during therapy.

The present study showed that companion and household animals (dogs, cats, sheep and goats) may act as a carrier of ESBL/biofilm-producing, multi-drug resistant, high-risk clonal lineage of *Klebsiella* (CTX-M-15-producing). The prevalence was significantly more in companion animals than the household animals probably due to therapeutic exposure of third-generation cephalosporins. The study made the caution about judicious use of antibiotics especially in companion animals.

## Materials and Methods

### Sampling

In total, 624 rectal swabs were collected from apparently healthy dogs (*n* = 254), cats (*n* = 108), sheep (*n* = 119) and goats (*n* = 143) of either sex and various age groups in Kolkata metropolitan, the cities and villages in Malda, Howrah, Nadia, South 24 Parganas and Jhargram districts of West Bengal during 2018–2019 (Table 1). The sampled dogs belonged to Labrador Retriever, Golden Retriever,

**Table 3** Antimicrobial resistance pattern of ESBL-producing *Klebsiella* isolated from companion and household animals in West Bengal, India

| Antimicrobial agents used              | Sensitive  | Intermediate | Resistant  |
|----------------------------------------|------------|--------------|------------|
| Ampicillin (10 µg)                     | 0          | 11 (11.8%)   | 82 (88.1%) |
| Ampicillin/cloxacillin (10/10 µg)      | 0          | 17 (18.2%)   | 76 (81.7%) |
| Amoxicillin/clavulanic acid (20/10 µg) | 9 (9.6%)   | 17 (18.2%)   | 67 (72.0%) |
| Cefotaxime (30 µg)                     | 20 (21.5%) | 26 (27.9%)   | 47 (50.5%) |
| Cefotaxime/clavulanic acid (30/10 µg)  | 7 (7.5%)   | 8 (8.6%)     | 78 (83.8%) |
| Ceftazidime (30 µg)                    | 38 (40.8%) | 11 (11.8%)   | 44 (47.3%) |
| Ceftazidime/clavulanic acid (30/10 µg) | 11 (11.8%) | 20 (21.5%)   | 62 (66.6%) |
| Ceftriaxone (30 µg)                    | 20 (21.5%) | 0            | 73 (78.4%) |
| Gentamicin (10 µg)                     | 40 (43.0%) | 2 (2.1%)     | 51 (54.8%) |
| Amikacin (30 µg)                       | 38 (40.8%) | 9 (9.6%)     | 46 (49.4%) |
| Ciprofloxacin (5 µg)                   | 5 (5.3%)   | 20 (21.5%)   | 68 (73.1%) |
| Tetracycline (30 µg)                   | 0          | 0            | 93 (100%)  |
| Co-trimoxazole (25 µg)                 | 0          | 8 (8.6%)     | 85 (91.3%) |
| Chloramphenicol (30 µg)                | 0          | 0            | 93 (100%)  |
| Azithromycin (15 µg)                   | 24 (25.8%) | 3 (3.2%)     | 66 (70.9%) |
| Clindamycin (2 µg)                     | 28 (30.1%) | 0            | 65 (69.8%) |
| Moxifloxacin (5 µg)                    | 26 (27.9%) | 18 (19.3%)   | 49 (52.6%) |
| Doxycycline (30 µg)                    | 0          | 8 (8.6%)     | 85 (91.3%) |
| Imipenem-EDTA (10/750 µg)              | 93 (100%)  | 0            | 0          |
| Piperacillin/tazobactam (100/10 µg)    | 34 (36.5%) | 5 (5.3%)     | 54 (58.0%) |
| Norfloxacin (10 µg)                    | 5 (5.3%)   | 11 (11.8%)   | 77 (82.7%) |

German Shepherd, Beagle, Pug, Saint Bernard, Spitz, Dachshund, Mongrel, Mudhol Hound and Rajapalayam breeds and the sheep and goats belonged to Garole and Black Bengal goats, respectively.

The dogs were kept mostly indoor, whereas the cats were allowed to roam around the houses by the animal owners. The practices of vaccination and visiting formal veterinarians during illness were found to be better for

companion animals than the household animals. The antibiotics prescribed in companion animals typically belonged to third- or fourth-generation cephalosporins such as cefixime (14.9%), ceftriaxone–tazobactam (12.5%), cephalexin (11.8%), ceftazidime (9.4%), cefotaxime (8.6%), cefoperazone–sulbactam (5.9%), doxycycline (11.0%), oxytetracycline (8.2%), enrofloxacin (7.0%), moxifloxacin (6.2%), ofloxacin (5.9%), amoxicillin–clavulanic acid (4.3%), clindamycin (3.9%) and azithromycin (3.5%; Table S1). The animal owners confessed about self-prescribing and stoppage of the antimicrobials before completion of the prescribed regime. In case of cats, the challenge of oral feeding of medications also influenced the owners to stop the antibiotics before completion of the course.

The studied household animals (sheep and goats) in villages were reared by semi-intensive system, which allowed roaming of the animals around the houses during the daytime. Mostly 1–2 animals were reared per household. There were no separate shelters or rooms for the animals and the animals had easy access to human cooking rooms, utensils, drinking water and food. The sheep and goats were offered with raw vegetables, cereals and kitchen leftovers. No commercial feed mixture was detected to be used for feeding. In spite of routine vaccination, the sheep and goats were occasionally treated during illness with oxytetracycline, enrofloxacin, lincomycin, amikacin, gentamicin and marbofloxacin mostly by paraveterinarians. The animal owners confessed about self-prescribing and stoppage of the antimicrobials when the symptoms subside which reduced their cost of treatment.

The rectal swabs were collected with the help of sterile cotton swab sticks (HiMedia, Mumbai, India) in sterile vials containing transport medium (HiMedia) as per the standard guideline. All the samples collected were brought into the laboratory maintaining the cold chain for further examination.

#### Isolation of *Klebsiella* and PCR-based confirmation

The samples collected in transport medium were inoculated into *Klebsiella* selective agar (HiMedia) and characteristic colonies were picked for further morphological and biochemical confirmation (Quinn *et al.* 1994). The chromogenic substrate incorporated in the selective media is cleaved specifically by *Klebsiella* and it produces a purple-magenta coloured colony. All the isolates were subjected to PCR for molecular confirmation as described earlier (Brisse and Verhoef 2001). All the PCR-positive *Klebsiella* isolates were further tested for the identification of *K. pneumoniae* by specific PCR although the PCR could not distinguish *K. pneumoniae* from other members of the *K. pneumoniae* species complex (Liu *et al.* 2008).

### Double disc diffusion test

PCR confirmed *Klebsiella* isolates were subjected to screening for extended spectrum beta-lactamase production by antibiotic susceptibility test containing cefotaxime (30 µg, HiMedia) and ceftazidime (30 µg, HiMedia) antibiotic discs with or without clavulanate (10 µg, HiMedia; CLSI 2017). Furthermore, ceftoxitin–cloxacillin double disc synergy (CC-DDS) was performed with all the isolates for phenotypic confirmation of ACBL production (Tan *et al.* 2009).

### Detection of beta-lactamase genes (*bla*<sub>CTX-M</sub>, *bla*<sub>TEM</sub>, *bla*<sub>SHV</sub>)

All the phenotypically ESBL-producing *Klebsiella* isolates including controls were subjected to PCR for detection of *bla*<sub>CTX-M</sub>, *bla*<sub>TEM</sub> and *bla*<sub>SHV</sub> genes (Cao *et al.* 2002; Weill *et al.* 2004). The selected PCR products were sequenced from commercially available sources (Xcelris Genomics, Ahmedabad, India). Sanger sequencing was conducted on ABI 3730xl 96 capillary system. The sequence homology searches were conducted using the BLAST algorithm (www.ncbi.nlm.nih.gov/BLAST). The selected sequences were submitted into DDBJ.

### Detection of chromosomal *bla*<sub>AmpC</sub>

All the phenotypically ACBL-producing *Klebsiella* isolates including controls were subjected to PCR for detection of *bla*<sub>AmpC</sub> (Féria *et al.* 2002). The selected PCR products were sequenced from commercially available sources (Xcelris Genomics). Sanger sequencing was conducted on ABI 3730xl 96 capillary system. The sequence homology searches were conducted using the BLAST algorithm (www.ncbi.nlm.nih.gov/BLAST).

### Detection of phenotypical biofilm production and biofilm-associated genes (*csgA*, *sdiA*, *rcaA*, *rpoS*)

The phenotypic detection of biofilm was conducted to all the isolates by tissue culture plate method (Mathur *et al.* 2006). The diluted culture of each isolate was inoculated into sterile, 96-well flat bottom, tissue culture plate (Tarsion, Kolkata, India). The plate was incubated at 37°C for 18 h. The content of each well was gently removed by tapping the plates and washed four times with PBS (pH 7.2) to remove free floating planktonic bacteria. Biofilm formed by the isolates was fixed with 2% sodium acetate and stained with 0.1% crystal violet. Excess stain was rinsed off with deionized water and the OD value was recorded at 570 nm with a microplate reader (EIAQuant, Meril, Valsad, India). Reference strain for biofilm-producing *Escherichia coli* (ATCC 35218, HiMedia) and

*Salmonella* (ATCC 35984, HiMedia) were used as positive controls. The method was conducted in triplicate and the OD values <0.12, 0.12–0.24 and >0.24 were considered as weak, moderate and high biofilm producers, respectively.

All the phenotypically biofilm-producing *Klebsiella* isolates including controls were subjected to PCR for detection of *csgA*, *sdiA*, *rcaA* and *rpoS* (Silva *et al.* 2013; Adamus-Bialek *et al.* 2015). The selected PCR products were sequenced from commercially available sources (Xcelris Genomics). Sanger sequencing was conducted on ABI 3730xl 96 capillary system. The sequence homology searches were conducted using the BLAST algorithm (www.ncbi.nlm.nih.gov/BLAST).

### Antimicrobial susceptibility of beta-lactamase-producing *Klebsiella* isolates

All the beta-lactamase-producing-*Klebsiella* isolates were tested for their phenotypical resistance against different antibiotics such as ampicillin (10 µg), ampicillin/cloxacillin (10/10 µg), amoxycillin/clavulanic acid (20/10 µg), cefotaxime (30 µg), cefotaxime/clavulanic acid (30/10 µg), ceftazidime (30 µg), ceftazidime/clavulanic acid (30/10 µg), ceftriaxone (30 µg), gentamicin (10 µg), amikacin (30 µg), ciprofloxacin (5 µg), co-trimoxazole (25 µg), tetracycline (30 µg), chloramphenicol (30 µg), azithromycin (15 µg), clindamycin (2 µg), moxifloxacin (5 µg), ceftoxitin (30 µg), doxycycline (30 µg), norfloxacin (10 µg), piperacillin/tazobactam (100/10 µg), imipenem-EDTA (10/750 µg) and colistin (10 µg) (HiMedia) (CLSI 2017). The interpretation for susceptibility testing (sensitive, intermediate, resistant) was conducted following the manufacturer chart prepared as per CLSI and EUCAST recommendation.

### Statistical analysis

Prevalence of beta-lactamase and biofilm-producing *Klebsiella* in companion and household animals was compared by chi-square test using SPSS software version 17.0 (SPSS Inc., Bangalore India). The *P* value was detected by the chi-square table to test a chi-square hypothesis. The small *P* value (<0.05) indicated the significant differences between the data and the rejection of null hypothesis.

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## Conflict of Interest

The authors declare no conflicts of interest.

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## Supporting Information

Additional Supporting Information may be found in the online version of this article:

**Table S1.** Antimicrobials usage in companion animals in Kolkata Metropolitan, India.